MJ23R2- designed and built in R.

Introduction:
The motivation behind MJ23R2 is to create a structured way to evaluate financial reporting discrepancies, debt settlement outcomes, operational risk patterns, and governance concerns within a single analytical framework.

I started with a simple question:
"How can we systematically identify when reported financial information begins to diverge from verified information and determine whether that divergence creates operational, settlement, legal, or governance risk?"
To answer that, I designed MJ23R2 as a multi-layer analytics model consisting of more than 30 interconnected scoring engines.

The framework begins by comparing reported versus verified financial data and calculating variance, incidence, consistency, and trend indicators.

These metrics serve as early warning signals that something may be changing within a portfolio, process, or reporting environment.

From there, the model moves into a structural intelligence layer where it evaluates reporting reliability, process integrity, remediation effectiveness, and operator-attributed risk.

Rather than simply identifying discrepancies, the framework attempts to determine whether observed patterns are more consistent with operational error, process control weaknesses, system issues, or elevated manipulation risk patterns.

The next layer focuses on settlement intelligence. I built a series of engines that analyze settlement achievement, efficiency, backlog concentration, and settlement distortion indicators. The intention is to identify situations where settlements may appear successful on the surface but still create underlying governance or exposure concerns.

The framework then incorporates litigation exposure indicators and escalation logic. Double-trigger and triple-trigger mechanisms increase risk severity when multiple adverse conditions occur simultaneously, such as rising variance, elevated incidence rates, and legal activity.

At the governance level, I created a composite metric called the Financial Integrity Index, or FII. This score aggregates variance exposure, reporting reliability, operational risk, and settlement distortion into a single governance-oriented indicator that can be used for portfolio prioritization and review.

One aspect I'm particularly proud of is the post-settlement re-exposure layer.

Most settlement models stop evaluating risk once a settlement is marked closed. I added a monitoring layer that detects when previously closed settlements become disputed, reopened, legally challenged, or financially invalidated.

The framework then recalculates risk indicators and governance priorities based on newly discovered information.

From my perspective analytics is often about creating a structured decision framework that transforms raw information into actionable insight.

Building MJ23R2 with industry in mind required me to think about both the technical implementation and the business decisions that the outputs were intended to support. This is why is chose to write the program exclusively in R with JSON comparable scripts that can easily be used with scalable large data sets without extra data consumption or nuance.